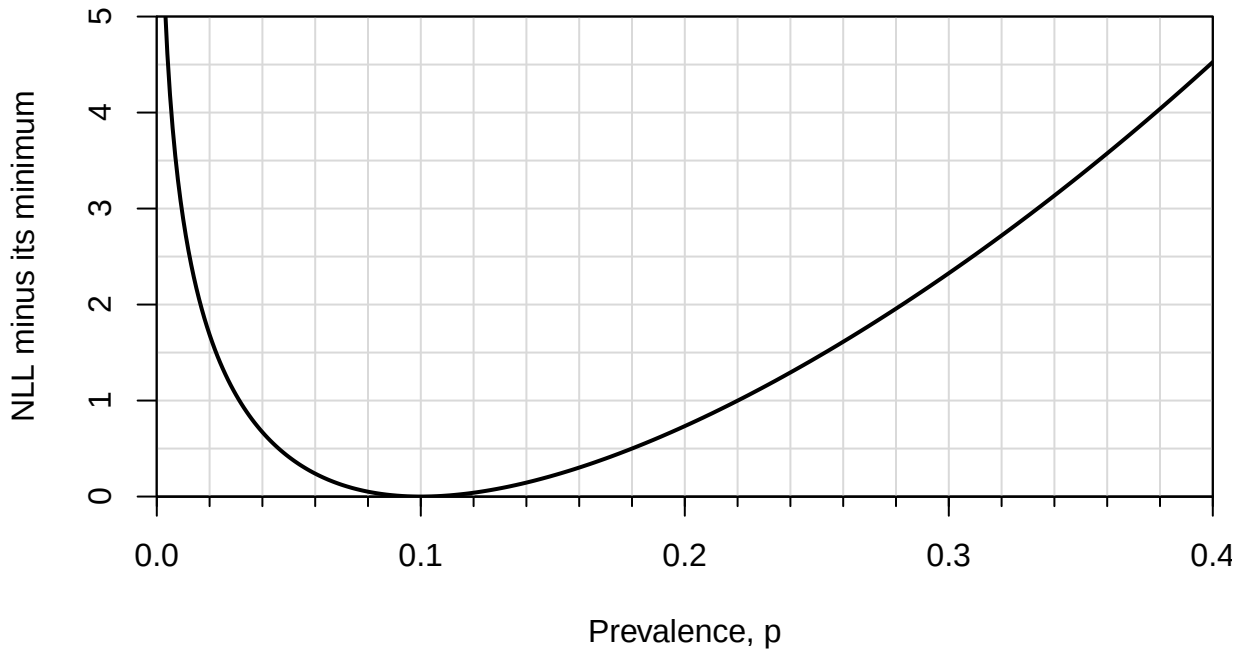
Answer key. Answers are in magenta.

NRES 746, Week 7, Lecture 1: Reading the Interval

Instructions: Work alone on Part 1 (about 4 minutes), then compare with a neighbor for Part 2. Keep laptops closed.

The data

You swab 20 frogs from a pond for chytrid fungus, and 2 test positive. Model the number of positives as binomial with unknown prevalence p. The figure shows the negative log-likelihood (NLL) for these data, minus its minimum value (so the lowest point of the curve is at zero).



Part 1: Read the curve (individual)

1a. What is the maximum likelihood estimate of p?

$\hat{p} \approx 0.10$

1b. A common rule of thumb treats every value of p whose NLL is within 2 units of the minimum as plausible. Draw that cutoff on the figure, then read off the approximate range of plausible values.

From 0.02 to 0.28

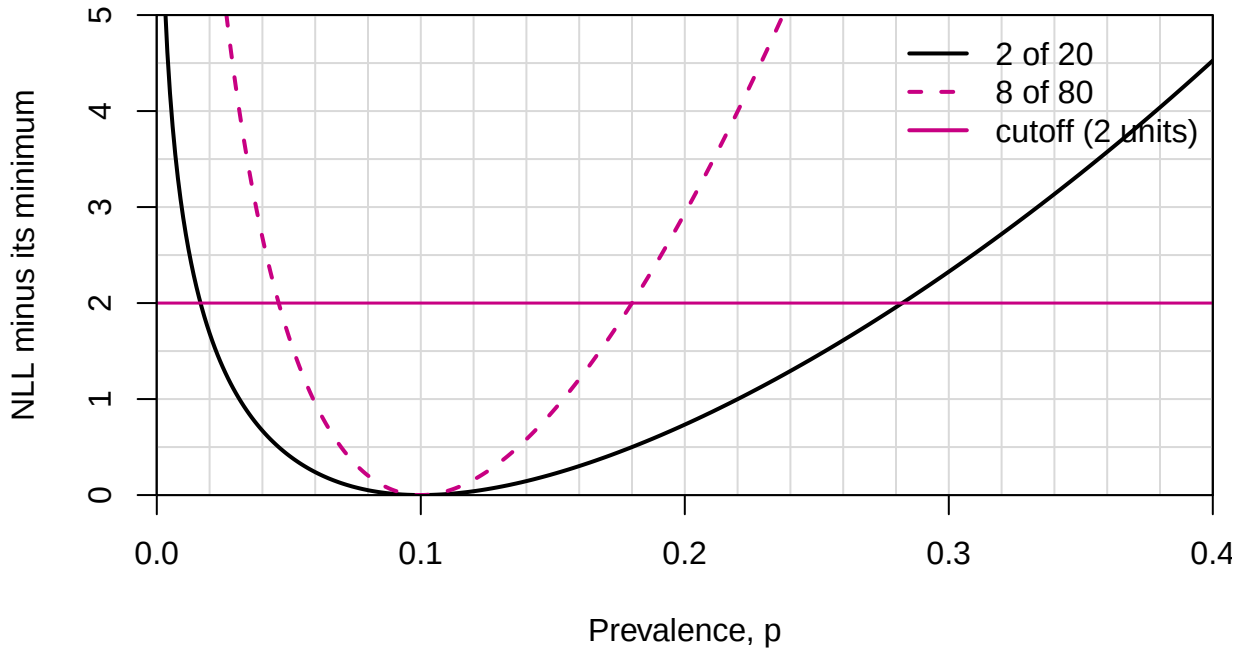
1c. Is this range *symmetric* around $\hat{p}$? Why might the curve be lopsided for these data?

No: it stretches farther above 0.10 than below. With only 2 positives the curve is squeezed against $p = 0$ (prevalence can't be negative), so it rises steeply on the left and slowly on the right.

Part 2: More frogs (with a neighbor)

2a. A second survey swabs 80 frogs and finds 8 positive. Sketch its curve on the same figure. Will its range of plausible values be wider or narrower than yours, and does $\hat{p}$ change? Compare sketches with your neighbor.

Narrower (about 0.05 to 0.18); $\hat{p}$ stays at 0.10. Four times the data makes the curve much steeper and less lopsided.


NLL minus its minimum
0
1
2
3
4
5
0.0
0.1
0.2
0.3
0.4
Prevalence, p
2 of 20
8 of 80
cutoff (2 units)